	Scotch Student ID #			
Circle the relevant digits	0	0	0	0
	1	1	1	1
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 4 – Organic Chemistry and Spectroscopic Analysis

Date: Monday August 15th, 2022

Reading Time	0 minutes
Writing Time	45 minutes

Task Sections	Marks	Your Marks
Total Marks	42	

General Instructions

- Write your student number in the space provided above on this page.
- Please complete all questions in the spaces provided.

Allowed Materials

- A VCAA Chemistry Data Book has been provided.
- Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.

At the end of the task…

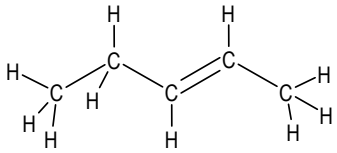
- Please ensure your student number has been filled in at the top of the page and that you have written your teacher's name.
- You may keep the data book.

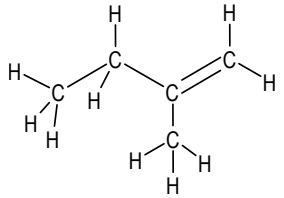
Electronic Devices

Students are **not** allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

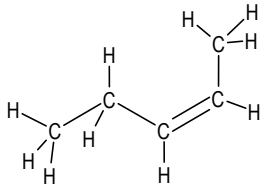
Q1.

Consider the structure at the right.
Which of the structures shown below does ***not*** represent an *isomer* of this?

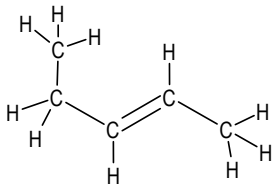




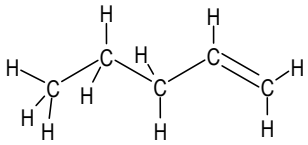
A



B



C



D

Q2.

Why is a trace of tetramethylsilane (TMS, $Si(CH_3)_4$) added to a sample before recording an NMR spectrum?

A It stabilises the sample to prevent chemical reactions occurring while in the spectrometer.

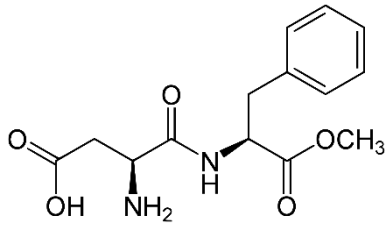
B It is an internal standard, producing a signal that is used as the basis for chemical shift measurement.

C It causes splitting of the signals, providing invaluable additional information about the structure.

D It allows the operator to check that the spectrometer is operating correctly.

Q3.

At the right is the structure of the artificial sweetener aspartame. Which of the following structural units are ***all present*** in the structure of this molecule?



A amino and ester group, and carboxy group

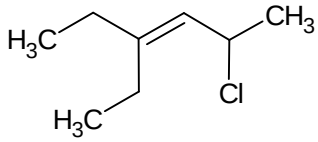
B ketone, carboxy, amino group

C amide and hydroxy group, and ester group

D carboxy, ketone, and amide groups

Q4.

The compound with the structure shown at the right would exhibit



A geometric (cis/trans) isomerism but not optical isomerism (chirality)

B optical isomerism (chirality) but not geometric (cis/trans) isomerism

C neither geometric (cis/trans) isomerism nor optical isomerism (chirality)

D both geometric (cis/trans) isomerism and optical isomerism (chirality)

Q5.

The balanced equation for hydroxylation of chlorethane using potassium hydroxide is

$CH_3CH_2Cl + KOH \rightarrow CH_3CH_2OH + KCl$

What effect would using LiOH instead of KOH have on the *atom economy* of this reaction?

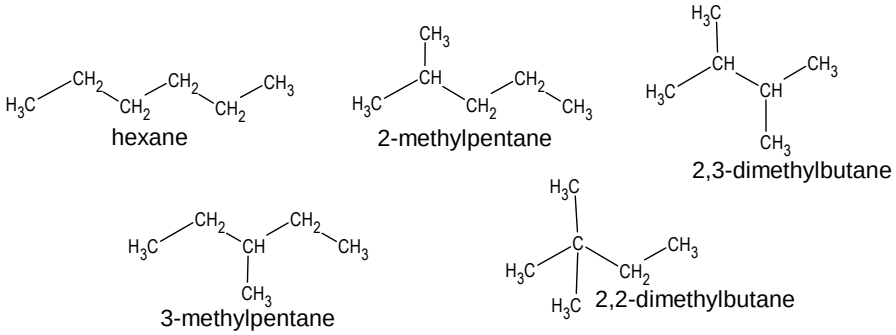
A The atom economy would be increased

B The atom economy would be decreased

C The atom economy would be unaffected

D Unknown. The effect on the atom economy must be determined experimentally

The five structural isomers of hexane shown here:



Q6. .

Of these, the ones that would show four signals in the ^{13}C NMR are

A 2-methylpentane and 3-methylpentane

B 3-methylpentane and 2,2-dimethylbutane

C 3-methylpentane and 2,3-dimethylbutane

D 2-methylpentane and hexane

Q7.

Butan-2-ol is oxidised using acidified potassium dichromate solution. Which of the following half equations correctly shows the oxidation of butan-2-ol?

A $C_4H_{10}O + H_2O \rightarrow C_4H_8O_2 + 4H^+ + 4e^-$

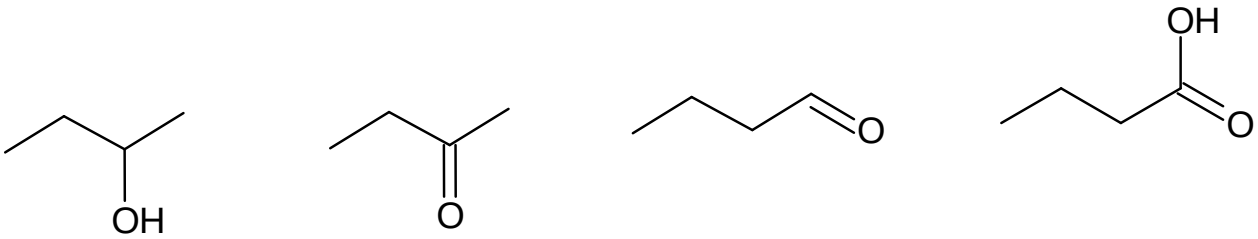
B $C_4H_{10}O \rightarrow C_4H_8O + 2H^+ + 2e^-$

C $C_4H_{10}O + H_2O \rightarrow C_4H_6O_2 + 6H^+ + 6e^-$

D $C_4H_{10}O \rightarrow C_4H_6O + 4H^+ + 4e^-$

Q8.

Which of the following molecules would produce a ^{13}C NMR spectrum including a signal with the ***highest*** chemical shift (δ) value?



A **B** **C** **D**

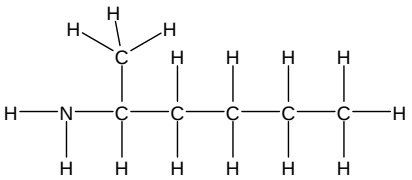
***Q9.* (4 marks)**

a) Which would have the higher boiling point, pentane (M=72.0 g mol^{-1}) or butan-1-ol (M=74.0 g mol^{-1})? Justify your prediction. (2 marks)

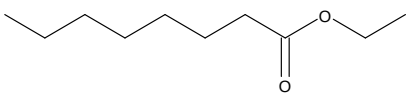
b) Which will be more soluble in water, hexanoic acid (M=116.0 g mol^{-1}) or octanoic acid (M=144.0 g mol^{-1})? Justify your prediction. (2 marks)

***Q10.* (4 marks)**

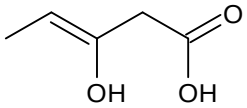
a) Give a systematic name for each of the structures shown below. (3 marks)



$CHC(CH_2)_3CH(CH_3)_2$

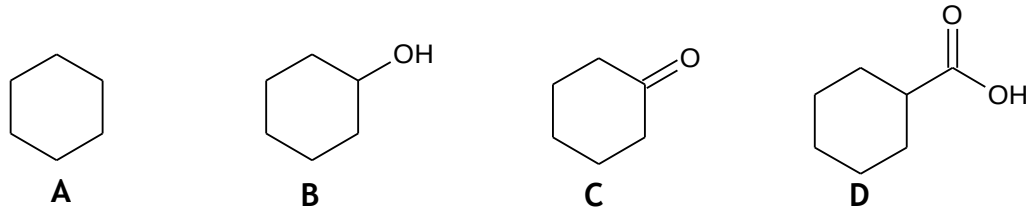


b) Write the semi-structural formula for the following structure. (1 mark)

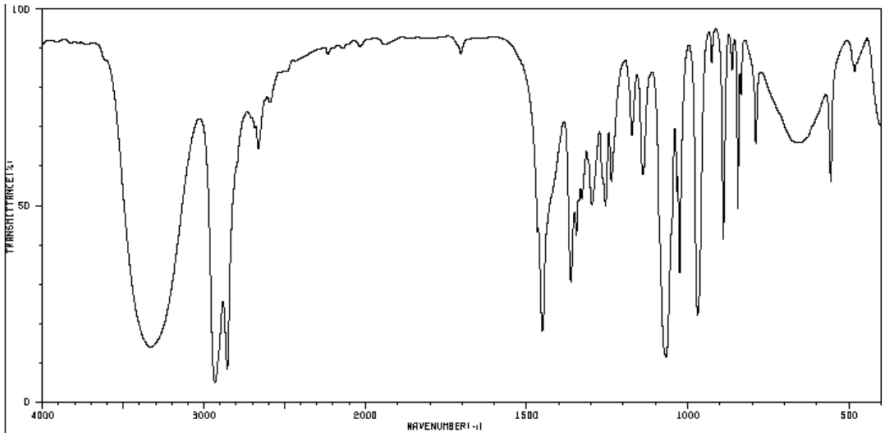


Q11. **(3 marks)**

Consider the following four structures.

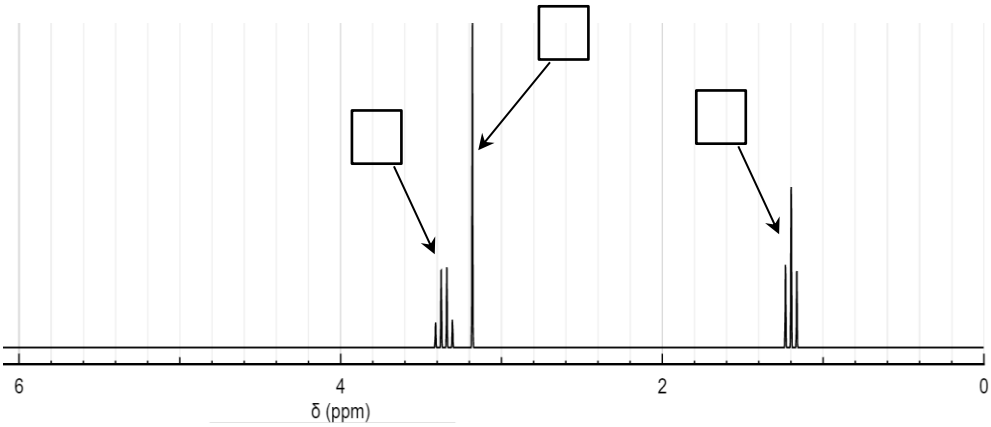


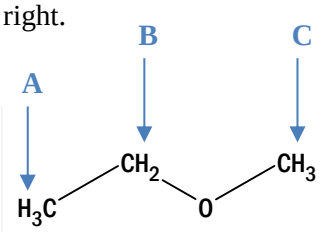
Which of these is most likely to have produced the following IR spectrum? Carefully justify your choice, by referring to all relevant conclusions that can be drawn from the spectrum. (3 marks)



Q12. **(6 marks)**

Below is the ^{1}H NMR spectrum of methoxyethane, whose structure is shown at the right.





δ (ppm)	*multiplicity*
1.2	triplet
3.2	singlet
3.4	quartet

a) The table gives details of the three signals recorded. Explain what the number of (low resolution) signals or peaks in the spectrum indicates about the structure. (1 mark)

b) In the boxes provided on the spectrum, clearly identify the hydrogen atoms in the structure (A, B, or C) responsible for each of the signals. (1 mark)

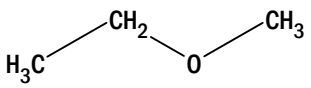
The molecule contains two methyl groups, labelled A and C.

c) Show how the ***chemical shifts in the Data Book*** can be used to distinguish between the signals produced by methyl groups A and C. (2 marks)

d) Explain how the ***multiplicities*** observed can be used to distinguish between the signals produced by methyl groups A and C. (2 marks)

Q13. **(7 marks)**

Methoxyethane is mixed with chlorine gas (Cl_2) and irradiated with UV light.



methoxyethane

The dominant reaction can be represented by the following equation:

$$C_3H_8O(g) + Cl_2(g) \xrightarrow{UV\ light} C_3H_7ClO(g) + HCl(g)$$

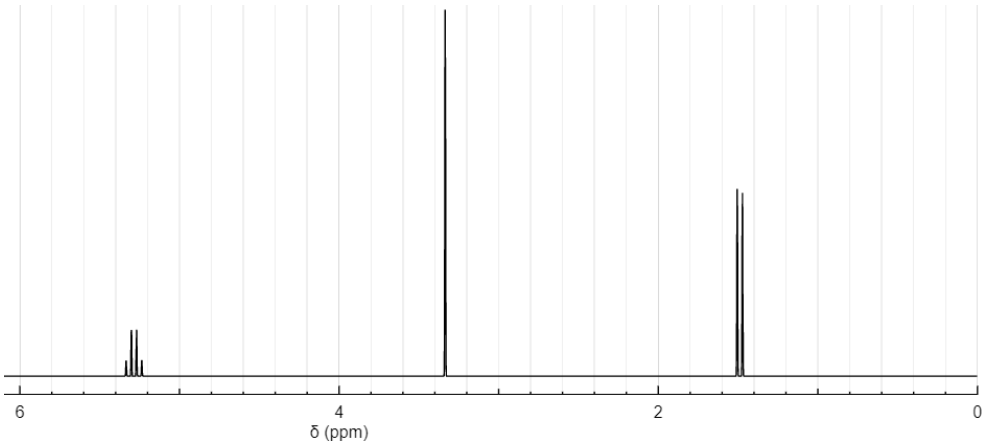
M(g mol^{-1}):	60.0	71.0	94.5	36.5

a) What ***type*** of reaction is this? (1 mark)

b) Calculate the atom economy of this reaction. (2 marks)

c) In one reaction 0.650 g of methoxyethane was reacted, and 0.450 g of the product, C_3H_7ClO, was isolated. Calculate the percentage yield. (2 marks)

d) The product, C_3H_7ClO, proved to be a mixture of structural isomers, which were separated by distillation. Below is the ^1H NMR spectrum of the predominant isomer obtained.



δ (ppm)	***multiplicity***
1.5	doublet
3.3	singlet
5.3	quartet

Draw the full structure of the product of the reaction that produced this spectrum. (2 marks)

Q14. **(4 marks)**

For each compound named below draw both a complete structure, ***showing all atoms and bonds***, and also a skeletal structure.

a) *cis*-but-2-enoic acid (2 marks)

full structure	skeletal structure

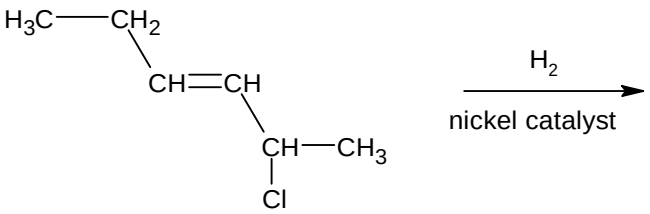
b) 3-amino-2,4-dichloropentan-2-ol (2 marks)

full structure	skeletal structure

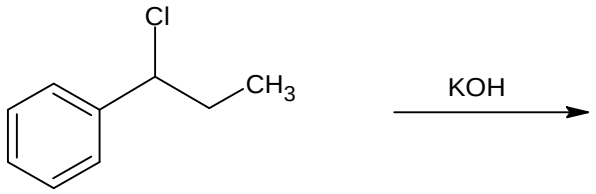
Q15. **(2 marks)**

In each of the following, draw a structure of the organic product formed (by-products need not be included).

a)

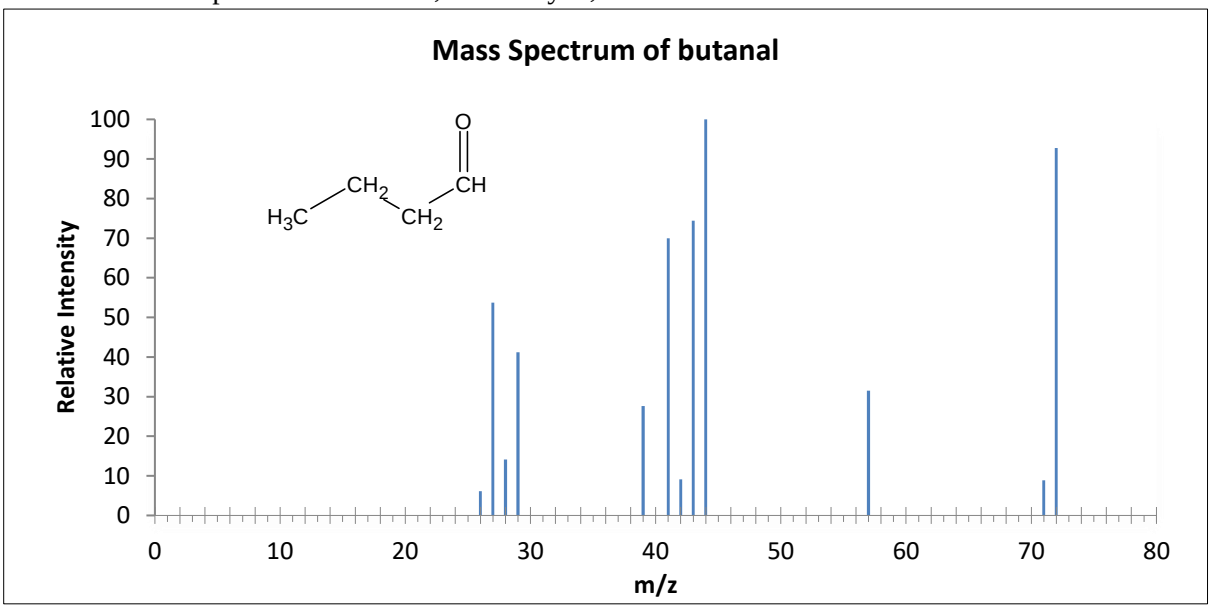


b)



Q16. **(4 marks)**

Below is a mass spectrum of butanal, an aldehyde, whose structure is also shown.



Refer to the spectrum and answer the following questions. (4 marks)

a) What are the m/z values for

i. the parent/molecular ion peak? .....

ii. the base peak? .....

b) The signals in the spectrum that are listed below can be attributed to fragments arising from simple fission (breaking) of one bond in the parent ion.
Write the formula of a particle formed this way that could be responsible for the signal at

i. m/z = 29 .....

ii. m/z = 57 .....

End of Task – 42 marks available